

Proposition I.3

To Cut Off from the Greater of Two Given Segments a Segment Equal to the Less

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1 Introduction

Euclid's third proposition asks for the construction of a segment equal to a given shorter segment on a given greater segment.

Classically, this construction is obtained by combining Proposition I.2 with an additional circle construction. A segment equal to the shorter one is first laid off from an endpoint of the greater segment, and a circle is then used to determine the required point on the greater segment.

In the present reconstruction over Hilbert geometry, the logical structure is substantially different. The theory developed in Book Zero already contains an abstract notion of one segment being less than another. This notion is defined by the existence of a point between the endpoints of the greater segment such that the corresponding subsegment is congruent to the smaller one.

Consequently, the existential content of Euclid's Proposition I.3 is already embodied in the definition of segment inequality. The reconstruction therefore consists not in repeating Euclid's geometric construction, but in exposing the witness contained in the definition of `HilbertSegmentLess`.

This proposition provides the first clear example where a classical Euclidean construction becomes an immediate consequence of the abstract language developed in Book Zero.

2 The Classical Configuration

Euclid's third proposition asks for the construction of a segment equal to a given shorter segment on a given greater segment.

Unlike Proposition I.2, the present construction does not introduce a new geometric technique. Instead, it combines Proposition I.2 with one additional circle construction.

The construction proceeds as follows.

2.1 Final Configuration

The objective is to determine a point E on the greater segment AB such that

$$AE \cong CD.$$

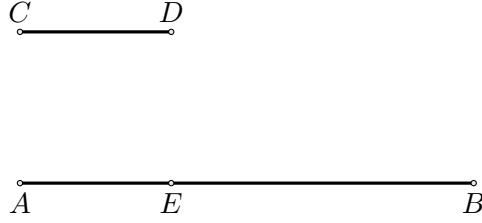


Figure 1: Final configuration of Euclid's construction for Proposition I.3.

The final geometric configuration is shown in Figure 1.

From the construction we obtain

$$AE \cong CD.$$

The remaining task is to justify that the point E lies between the endpoints of the greater segment.

2.2 Construction Algorithm

The construction uses Proposition I.2 as an already established construction primitive.

2.2.1 Step 1. Given Segments

Let AB be the greater segment and CD the given shorter segment.

2.2.2 Step 2. Apply Proposition I.2

By Proposition I.2, construct a segment AF congruent to the given segment CD .

Thus

$$AF \cong CD.$$

2.2.3 Step 3. Draw the Circle Centered at A

Draw the circle centered at A passing through F .

2.2.4 Step 4. Determine the Required Point

Let E be the intersection of the circle with the greater segment.

Since

$$AE \cong AF$$

and

$$AF \cong CD,$$

it follows that

$$AE \cong CD.$$

The required segment has therefore been obtained.

3 Classical Proof

The classical Euclidean proof constructs the required point by applying Proposition I.2 and then introducing an auxiliary circle.

The reconstruction over Hilbert geometry follows a different logical route. The relation that one segment is less than another is already represented in Book Zero by the predicate

$$\text{HilbertSegmentLess } CD \ AB.$$

By definition, this predicate means that there exists a point E between the endpoints of the greater segment satisfying

$$A - E - B$$

and

$$AE \cong CD.$$

Therefore, assuming

$$CD < AB,$$

we immediately obtain a point E such that

$$A - E - B \quad \text{and} \quad AE \cong CD.$$

This is precisely the conclusion required by Proposition I.3.

Consequently, over the present Hilbert/Book Zero layer, the proposition is an immediate consequence of the definition of `HilbertSegmentLess`.

□

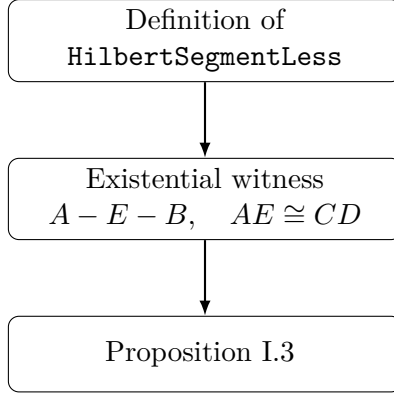


Figure 2: Logical structure of Proposition I.3 over the Book Zero layer.

4 Proof Architecture of the Reconstruction

5 Hilbert Reconstruction

Unlike Euclid’s original proof, the present reconstruction does not repeat the geometric construction based on Proposition I.2.

The abstract notion of one segment being less than another has already been introduced in Book Zero by the predicate `HilbertSegmentLess`. By definition, this predicate contains exactly the existential information required by Proposition I.3.

Consequently, the reconstruction consists simply in extracting the witness from this definition.

This illustrates an important feature of the Book Zero layer. Classical geometric constructions are progressively replaced by abstract synthetic notions whose mathematical content has already been established.

6 Lean Formalization

The Lean formalization closely follows the mathematical argument.

The hypothesis

$$\text{HilbertSegmentLess } CD \ AB$$

already contains an existential witness providing the required point between the endpoints of the greater segment together with the corresponding congruence.

The formal proof therefore consists only of unpacking this witness and applying the symmetry of segment congruence.

Unlike the classical Euclidean proof, no additional geometric construction is required.

The resulting Lean theorem is therefore considerably shorter than the corresponding classical argument.

7 Relationship with Earlier Propositions

7.1 Proposition I.2

Proposition I.3 is the first direct application of the segment-transfer construction established in I.2.

I.2 says that a segment congruent to a given segment can be placed at a prescribed point. I.3 adds an order constraint: when the target segment is longer than the given one, the transferred length can be realized as an initial part of that longer segment.

Thus I.3 converts free segment transport into controlled segment cutting.

7.2 Proposition I.1

Proposition I.1 enters indirectly through I.2. Euclid's segment-transfer construction uses equilateral triangles as part of the mechanism for moving a length to a new position.

By the time I.3 is reached, this construction has already been packaged as Proposition I.2 and can be used as a reusable operation.

7.3 Transition to Proposition I.4

Propositions I.1–I.3 form the initial construction block of Book I: equilateral triangle construction, segment transport, and segment cutting.

Proposition I.4 changes the character of the development. It is no longer primarily a construction theorem but a rigidity theorem for triangles: SAS.

Thus I.3 closes the first elementary segment-construction sequence and prepares the congruence theory beginning with I.4.

8 Hilbert and Book Zero Components Used

8.1 Strict Segment Comparison

The hypothesis that one segment is greater than another is represented synthetically by the established strict segment-order relation rather than by numerical lengths.

This relation carries the geometric witness needed to locate the shorter length inside the longer segment.

8.2 Segment Layoff

The Hilbert congruence infrastructure supplies a point at a prescribed distance along the relevant ray. This is the formal analogue of Euclid's use of Proposition I.2.

8.3 Betweenness

The conclusion of I.3 is not merely the existence of a congruent segment. The new endpoint must lie between the endpoints of the longer segment.

The Hilbert order layer records this positional requirement explicitly.

8.4 Congruence Transport

Segment congruence symmetry and transitivity are used to convert the constructed layoff segment into exactly the orientation of congruence required by the Euclidean statement.

9 Formalization Notes

Proposition I.3 is mathematically simple but formally instructive.

In modern numerical language one might describe the theorem as follows: if

$$|AB| > |CD|,$$

then there is a point E on AB such that

$$|AE| = |CD|.$$

The reconstruction deliberately avoids this numerical formulation. “Greater than” is represented by a synthetic segment-order relation, and the conclusion consists of two separate geometric facts:

$$A - E - B \quad \text{and} \quad AE \cong CD.$$

This distinction is important. Congruence alone would allow a copy of CD to be placed anywhere; Proposition I.3 requires the copy to occur inside the prescribed longer segment.

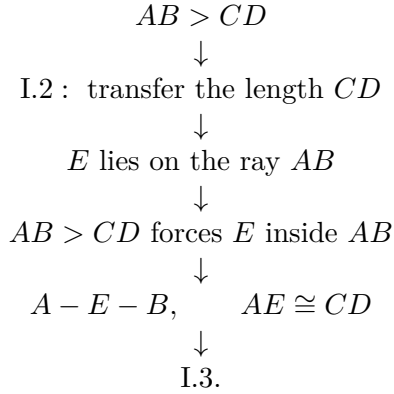
The formal theorem therefore combines the two foundational layers that will recur throughout Book I:

$$\text{congruence} \quad + \quad \text{order}.$$

In the Hilbert reconstruction the actual segment-layoff operation is already available at the interface level. Consequently the public Euclidean theorem is relatively short: its main role is to package layoff together with the betweenness condition dictated by the strict segment comparison.

10 Dependency Summary

The classical dependency pattern is



The formal reconstruction expresses the same structure through strict segment comparison, segment layoff, betweenness, and congruence transport.

No numerical length function or coordinate system is introduced.

11 Conclusion

Proposition I.3 completes the first elementary segment-construction block of Book I.

Proposition I.2 established free transport of a segment length to a prescribed point. Proposition I.3 adds order: the copied segment is cut from a longer prescribed segment and its endpoint must lie between the original endpoints.

The formalization makes this distinction particularly clear. The theorem is not simply about equality of lengths; it is about the interaction of segment congruence with betweenness.

This combination of congruence and order becomes fundamental throughout the later development. I.3 therefore serves as a compact early example of the two Hilbert layers working together before Book I turns in Proposition I.4 to the rigidity theory of triangles.